Insights and opportunities for deep brain stimulation as a brain circuit intervention

Paper: Neumann, Horn and Kuehn (2023). Review.

DOI: 10.1016/j.tins.2023.03.009

Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Measurement, stimulation targets, symptom mechanisms and feedback objectives. Connects local physiology with whole-network DBS effects.

How does it answer the general questions?

Measured and controlled. The review connects unit activity, LFP oscillations, cortical signals and imaging with stimulation of neural circuits. DBS can change synchrony and symptoms through distributed pathways; the effect depends on anatomy and stimulation settings (PDF pp. 1–8).

Usable insight. Connectomic targeting and physiological feedback address complementary spatial and temporal aspects. Adaptive stimulation can use symptom-linked biomarkers, but the review explicitly leaves their causal role open. Physiological and pathological synchrony must be distinguished before selecting an objective (PDF pp. 5–6, 12).

Model relevance. Supports choosing measurement and actuation channels around an explicit symptom/network hypothesis. Tract correlations alone cannot establish whether a pathway is activated or inhibited, or the direction of propagation. The review supplies neither a single fitted delayed rate model nor a PIE multiplier construction (PDF pp. 8–12).

What papers does it cite that could provide more information?

- Little et al. (2013), *Adaptive deep brain stimulation in advanced Parkinson disease*: early biomarker-driven intervention (ref. 73, PDF p. 14).
- Gradinaru et al. (2009), *Optical deconstruction of parkinsonian neural circuitry*: pathway-specific causal interrogation (ref. 147, PDF p. 16).